

PHP Object-Oriented Programming Assignment

Assignment Overview

In this assignment, you will design a small PHP system for an **Online Learning Platform**. Your goal is to demonstrate a clear understanding of object-oriented programming concepts covered in class.

Abstract Class Requirements

Create an abstract class named **User**.

Properties

- **private** property: *id* (stores the user ID)
- **protected** property: *name* (stores the user name)

Methods

- **Concrete Method:** *displayUserInfo()*
 - Displays the user's ID and name.
 - This method must be inherited by child classes.
- **Abstract Method:** *getRole()*
 - Returns the role of the user (example: Student, Instructor).
 - Each child class must implement this method differently.

Child Class Requirements (Inheritance)

Create at least **two child classes** that inherit from *User*.

Required Classes

- **Student**
- **Instructor**

Method Behavior

- Both classes must implement the *getRole()* method.
- Each class should return a role name that clearly identifies the user type.
- Child classes may add their own additional methods if needed.

Interface Requirements

Create an interface named **Authenticatable**.

Interface Methods

- **login()**
 - Displays a message indicating that the user has logged in.
- **logout()**
 - Displays a message indicating that the user has logged out.

At least one class must implement this interface.

Polymorphism Requirement

- Add an instance of class *Student* Use a reference of the abstract class type (*User*) to store objects of child classes.
- Call the *getRole()* method using this reference.

Output Requirements

- Output must clearly show:
 - User information
 - User role
 - Login and logout actions
- Messages should clearly indicate which class is responsible for the output.

Important Rules:

- Do not use PHP frameworks.
- Do not place all code in one file.